

PROJECT SUBMISSION

Medical Laboratory System — Project 4

1. Project Links

GitHub Repository	https://github.com/malakallam6/Medical-Laboratory-System.git
Presentation video link	https://nileuniversity-my.sharepoint.com/:f:/g/personal/m_ali2407_nu_edu_eg/IgAC0EPu9iDRSYhc5iBhE-UgAWjmO9F_4yWZzu6_EbV_YEQ?e=KwqnkH
Demo Link	https://nileuniversity-my.sharepoint.com/:v:/g/personal/m_ali2407_nu_edu_eg/IQC3onzkzT5lQr9jli9kAfNfARO4HIPEiZqCRcQycrKFfwA?nav=eyJyZWZlcnJhbEluZm8iOnsicmVmZXJyYWxBcHAIoiJPbmVEcmI2ZUZvckJ1c2luZXNzliwicmVmZXJyYWxBcHBQbGF0Zm9ybSI6IldlYiIsInJlZmVycmFsTW9kZSI6InZpZXciLCJyZWZlcnJhbFZpZXciOiJNeUZpbGVzTGlua0NvcHkifX0&e=h4YSy0

2. Team Members

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3. Project Report

Medical Laboratory System

Project Summary

Designed and implemented a relational database for a Medical Laboratory System that tracks laboratorians, patients, medical tests, test results, and lab components. The system includes 10 tables, 13 INSERT rows for TEST_RESULT, a stock-decrement trigger, a composite index for low-stock alerts, and a Python/Tkinter desktop UI connected to MySQL.

Key Features & Components

Database Tables 10 relational tables	TEST_RESULT Inserts 13 INSERT rows
Trigger Stock-decrement trigger	Indexing Composite index for low-stock alerts
Frontend / UI Python/Tkinter desktop application connected to MySQL	

Entities Tracked

The system manages data for the following entities: Laboratorians, Patients, Medical Tests, Test Results, and Lab Components — providing full traceability from patient registration through test execution to result reporting and inventory management.
